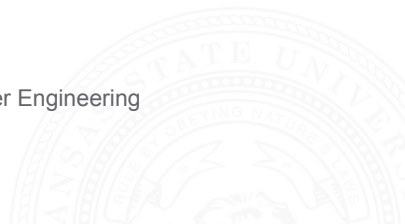


Three-Phase AC Circuits I

Balanced sets, phase sequence, and wye/delta relations

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Lecture outline

- 1 Balanced sets and phase sequence
- 2 Symmetrical components
- 3 Wye connection
- 4 Delta connection
- 5 Worked example

All voltage and current phasor magnitudes are RMS.

PART 1

Balanced sets and phase sequence

Phase labels, phasor angles, and waveform peak order

Why three phases: constant total power

Balanced sinusoidal set: RMS V_p, I_p ; common load angle ϕ .

For $\delta_k = 0^\circ, 120^\circ, 240^\circ$:

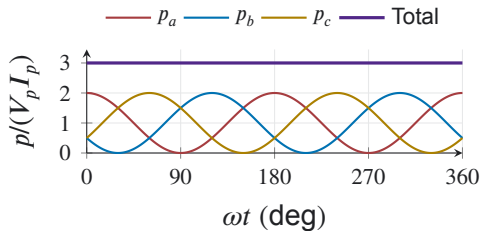
$$p_k = v_k i_k = V_p I_p \cos \phi + V_p I_p \cos(2\omega t - 2\delta_k - \phi).$$

The three 2ω terms cancel:

$$\sum_{k=a,b,c} \cos(2\omega t - 2\delta_k - \phi) = 0.$$

Only the constant terms remain:

$$p_a + p_b + p_c = 3V_p I_p \cos \phi.$$



For $\phi = 0^\circ$, powers in units of $V_p I_p$:

ωt	p_a	p_b	p_c	Total
0°	2	0.5	0.5	3
90°	0	1.5	1.5	3

Phase-power pulsations cancel; the total instantaneous power stays constant.

Why three phases: economical conductors

For balanced sinusoidal currents,

$$i_a = \hat{I} \cos \omega t,$$

$$i_b = \hat{I} \cos(\omega t - 120^\circ),$$

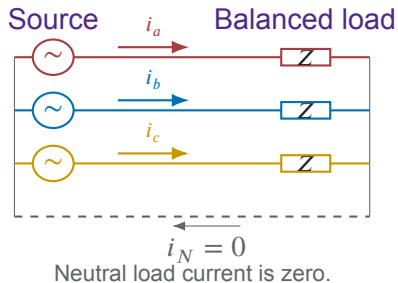
$$i_c = \hat{I} \cos(\omega t - 240^\circ).$$

At the common neutral, KCL gives

$$i_N = i_a + i_b + i_c = 0.$$

Separate returns: six wires.

Shared balanced return: three wires.



Three line resistances $R = \rho \ell / A$:

$$P_{\text{loss}} = 3I_{\text{rms}}^2 R \quad \text{versus} \quad 6I_{\text{rms}}^2 R$$

Same current and wire; three lines versus three pairs.

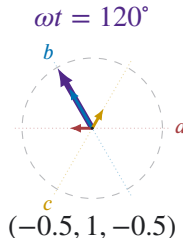
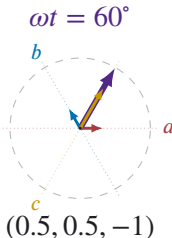
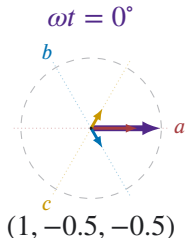
The other phases provide the return path; balanced load current needs no neutral.

Why three phases: a rotating magnetic field

Three winding axes are 120° apart in space; their currents are 120° apart in time.

Let $\beta_k = 0^\circ, 120^\circ, 240^\circ$, $i_k = \hat{I} \cos(\omega t - \beta_k)$, and $\hat{B}$ be one phase's peak field.

$$\mathbf{B}_k = \hat{B} \cos(\omega t - \beta_k)(\cos \beta_k \hat{\mathbf{x}} + \sin \beta_k \hat{\mathbf{y}}) \implies \mathbf{B}_{\text{total}} = \frac{3}{2} \hat{B} (\cos \omega t \hat{\mathbf{x}} + \sin \omega t \hat{\mathbf{y}}).$$



Currents below each plot: $(i_a, i_b, i_c)/\hat{I}$. Colored: phase fields. Purple: sum.

The resultant field keeps the same magnitude while its direction rotates, driving motor torque.

Balanced source

Balanced source voltages

- 1 one common sinusoidal frequency
- 2 three equal RMS magnitudes
- 3 exactly 120° between adjacent phase angles.

Balanced network

- identical phase-path impedances.
- identical phase loads.
- the same connection pattern in each phase.

A balanced voltage set describes the source. Connected loads and lines require separate balance checks.

Phasor-to-waveform mapping

$$\mathbf{V}_{xn} = V_{\text{rms}} \angle \theta_x \quad \Longleftrightarrow \quad v_{xn}(t) = \sqrt{2} V_{\text{rms}} \cos(2\pi f t + \theta_x)$$

Phasor record

Magnitude V_{rms} . RMS value stored in either polar or rectangular form.

Angle θ_x . Cosine phase relative to the declared reference.

Time-domain record

Peak $\hat{V} = \sqrt{2} V_{\text{rms}}$. Time-domain coefficient.

Common rate $\omega = 2\pi f$. Suppressed in the phasor.

Below, the shorthand V means V_{rms} ; all three phase phasors use the same frequency and angle reference.

Sequence rule

Positive sequence - abc

$$V_{an} = V \angle \theta, \quad V_{bn} = V \angle (\theta - 120^\circ),$$

$$V_{cn} = V \angle (\theta + 120^\circ).$$

Peak order: **a**, then **b**, then **c**.

Negative sequence - acb

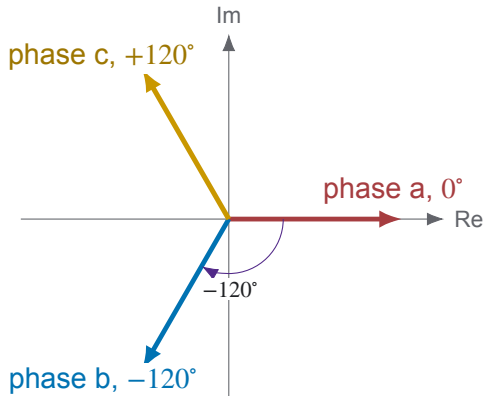
$$V_{an} = V \angle \theta, \quad V_{cn} = V \angle (\theta - 120^\circ),$$

$$V_{bn} = V \angle (\theta + 120^\circ).$$

Peak order: **a**, then **c**, then **b**.

Equal magnitude and spacing establish balance. The b/c positions establish the sequence.

Positive abc geometry



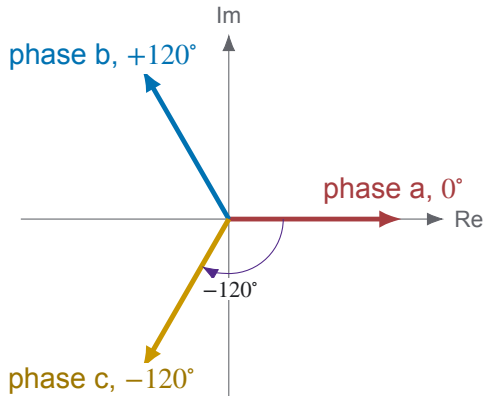
Peak order

Phase a is at its positive peak now. As the vectors rotate counterclockwise, phase b reaches the positive real axis next, followed by phase c:

$$a \rightarrow b \rightarrow c.$$

Each phase reaches its positive peak when its rotating vector points along the positive real axis.

Negative acb geometry



Peak order

Phase a is at its positive peak now. As the vectors rotate counterclockwise, phase c reaches the positive real axis next, followed by phase b:

$$a \rightarrow c \rightarrow b.$$

The vectors still rotate counterclockwise. Swapping b and c reverses the peak order.

Phase angles and peak times

Use the same cosine convention as L05. At $f = 60$ Hz,

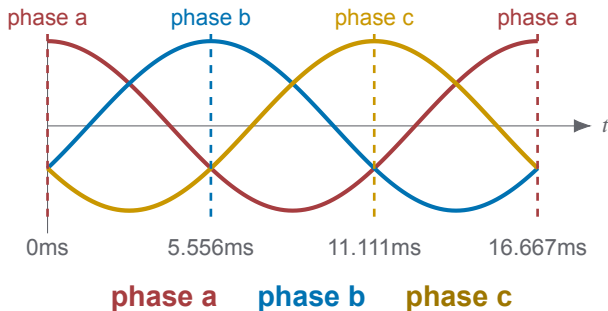
$$v_{an}(t) = \sqrt{2}V \cos(\omega t),$$

$$v_{bn}(t) = \sqrt{2}V \cos(\omega t - 2\pi/3), \quad \omega = 120\pi \text{ rad/s.}$$

$$v_{cn}(t) = \sqrt{2}V \cos(\omega t + 2\pi/3),$$

A positive peak occurs when the cosine argument equals $2\pi k$, for integer k .

Time-order test



First positive peaks after the phase-a peak

With $t_a = 0$ and $f = 60$ Hz,

$$t_b = \frac{2\pi/3}{120\pi} = \frac{1}{180} \text{ s} \\ \approx 5.556 \text{ ms},$$

$$t_c = \frac{4\pi/3}{120\pi} = \frac{1}{90} \text{ s} \\ \approx 11.111 \text{ ms}.$$

Positive peaks occur in the order $a \rightarrow b \rightarrow c$, with spacing $T/3 \approx 5.556$ ms.

Three forms of a complex number

One complex number z ; three equivalent ways to write it. Here $j^2 = -1$.

Rectangular

$$z = x + jy$$

Real and imaginary parts

Polar

$$z = r\angle\theta$$

Magnitude and angle

Exponential

$$z = re^{j\theta}$$

Magnitude times a rotation

Euler's formula links the forms (θ in radians):

$$e^{j\theta} = \cos \theta + j \sin \theta$$

$\implies$

$$re^{j\theta} = r \cos \theta + jr \sin \theta.$$

All three forms can be used to analyze three-phase power systems.



Balanced phasor sum

Unit phasors at 0° and $\pm 120^\circ$

$$1 + e^{j2\pi/3} + e^{-j2\pi/3} = 0.$$

For any reference angle θ , multiply by $V e^{j\theta}$:

$$V_{an} + V_{bn} + V_{cn} = 0.$$

A common rotation changes each phasor, but preserves their zero sum.

Component check at $\theta = 0^\circ$

Equivalently, $\theta = 2\pi m$ radians for integer m .

Real parts

$$V - \frac{1}{2}V - \frac{1}{2}V = 0$$

Horizontal components cancel.

Imaginary parts

$$0 - \frac{\sqrt{3}}{2}V + \frac{\sqrt{3}}{2}V = 0$$

Vertical components cancel.

For equal-frequency sinusoids, the zero phasor sum gives $v_a(t) + v_b(t) + v_c(t) = 0$ at every instant.

Example: positive-sequence phasors

Given $V_{an} = 120\angle 0^\circ$ V RMS and $f = 60$ Hz, construct positive abc :

$$V_{an} = 120\angle 0^\circ = 120 + j0,$$

$$V_{bn} = 120\angle(-120^\circ) = -60 - j60\sqrt{3},$$

$$V_{cn} = 120\angle 120^\circ = -60 + j60\sqrt{3} \text{ V}.$$

$$V_{an} + V_{bn} + V_{cn} = (120 - 60 - 60) + j(-60\sqrt{3} + 60\sqrt{3}) = 0.$$

The b and c labels fix positive sequence. Exact rectangular components make the zero sum visible.

Example: negative-sequence phasors

Keep $V_{an} = 120\angle 0^\circ$ V and exchange the b and c assignments:

$$V_{an} = 120 + j0,$$

$$V_{bn} = 120\angle 120^\circ = -60 + j103.923,$$

$$V_{cn} = 120\angle (-120^\circ) = -60 - j103.923 \text{ V}.$$

Balance

All magnitudes remain 120 V and the phasor sum remains zero.

Future positive peaks

$t_a = 0$, $t_c = T/3 = 5.556$ ms, and
 $t_b = 2T/3 = 11.111$ ms.

The negative sequence is $a \rightarrow c \rightarrow b$. Equal magnitudes and zero sum do not distinguish the two sequences.

Example: instantaneous voltage checks

For the positive abc set, let $\hat{V} = 120\sqrt{2} = 169.706$ V. At $f = 60$ Hz,

$$T = \frac{1}{f} = 16.667 \text{ ms}, \quad \omega = 2\pi f = \frac{2\pi}{T} = 120\pi \text{ rad/s}, \quad \boxed{\omega t = \frac{2\pi t}{T}}.$$

$$v_a = \hat{V} \cos \omega t, \quad v_b = \hat{V} \cos(\omega t - 2\pi/3), \quad v_c = \hat{V} \cos(\omega t + 2\pi/3).$$

Time	$v_a/\hat{V}$	$v_b/\hat{V}$	$v_c/\hat{V}$	Sum
0	1	-1/2	-1/2	0
$T/3 = 5.556 \text{ ms}$	-1/2	1	-1/2	0
$2T/3 = 11.111 \text{ ms}$	-1/2	-1/2	1	0

At each positive peak the other two phase voltages are -84.853 V. Exchanging columns b and c gives the negative-sequence timing.

Shifted reference

Common +25° shift

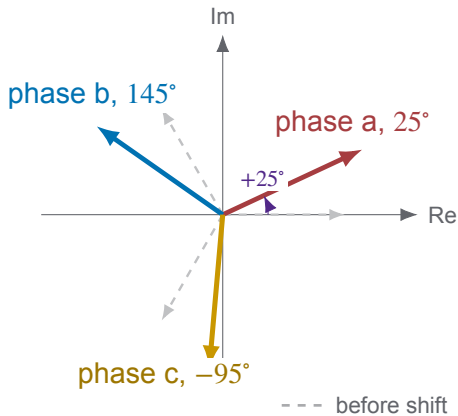
$$\begin{aligned} V_{an} &= 120\angle(0^\circ + 25^\circ) \\ &= 120\angle 25^\circ, \end{aligned}$$

$$\begin{aligned} V_{bn} &= 120\angle(120^\circ + 25^\circ) \\ &= 120\angle 145^\circ, \end{aligned}$$

$$\begin{aligned} V_{cn} &= 120\angle(-120^\circ + 25^\circ) \\ &= 120\angle(-95^\circ) \text{ V.} \end{aligned}$$

$$V_{an} + V_{bn} + V_{cn} = 0$$

After rotation



PART 2

Symmetrical components

An unbalanced set as positive, negative, and zero sequence

Three sequence basis vectors

Define the 120° rotation operator $a = e^{j120^\circ} = 1\angle 120^\circ$, so $a^2 = e^{-j120^\circ} = 1\angle(-120^\circ)$.

Three basis vectors

Positive abc

$$\mathbf{b}_1 = \begin{bmatrix} 1 \\ a^2 \\ a \end{bmatrix}$$

$$V_1 \equiv V_{a1}$$

Negative acb

$$\mathbf{b}_2 = \begin{bmatrix} 1 \\ a \\ a^2 \end{bmatrix}$$

$$V_2 \equiv V_{a2}$$

Zero, in phase

$$\mathbf{b}_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$V_0 \equiv V_{a0}$$

With entries ordered as (a, b, c) , $\mathbf{b}_1$ has angles $(0^\circ, -120^\circ, +120^\circ)$ and $\mathbf{b}_2$ has $(0^\circ, +120^\circ, -120^\circ)$. The first entry is 1, so V_1 , V_2 , and V_0 are the phase-a references.

Every three-phase phasor vector at one frequency has the unique decomposition

$$\mathbf{V}_{abc} = V_1 \mathbf{b}_1 + V_2 \mathbf{b}_2 + V_0 \mathbf{b}_0.$$

Why do the weighted sums isolate one sequence?

From the basis expansion

$$V_{an} = V_1 + V_2 + V_0,$$

$$V_{bn} = a^2 V_1 + a V_2 + V_0,$$

$$V_{cn} = a V_1 + a^2 V_2 + V_0.$$

Rotation identities

$$a^3 = 1, \quad 1 + a + a^2 = 0.$$

Example: isolate the positive sequence

$$\begin{aligned} V_{an} + a V_{bn} + a^2 V_{cn} &= \underbrace{(1 + a^3 + a^3)}_3 V_1 \\ &\quad + \underbrace{(1 + a^2 + a^4)}_0 V_2 + \underbrace{(1 + a + a^2)}_0 V_0 \\ &= 3V_1, \end{aligned}$$

because $a^4 = a$. Therefore

$$V_1 = \frac{1}{3} (V_{an} + a V_{bn} + a^2 V_{cn}).$$

The desired sequence adds to $3V_1$; the other two cancel because $1 + a + a^2 = 0$.

Finding the three sequence coefficients

Start from $\mathbf{V}_{abc} = V_1 \mathbf{b}_1 + V_2 \mathbf{b}_2 + V_0 \mathbf{b}_0$. Solving for the three scalar coefficients gives:

$$V_1 = \frac{1}{3} (V_{an} + aV_{bn} + a^2V_{cn}) ,$$

$$V_2 = \frac{1}{3} (V_{an} + a^2V_{bn} + aV_{cn}) ,$$

$$V_0 = \frac{1}{3} (V_{an} + V_{bn} + V_{cn}) .$$

V_1 scales $\mathbf{b}_1$ positive sequence

V_2 scales $\mathbf{b}_2$ negative sequence

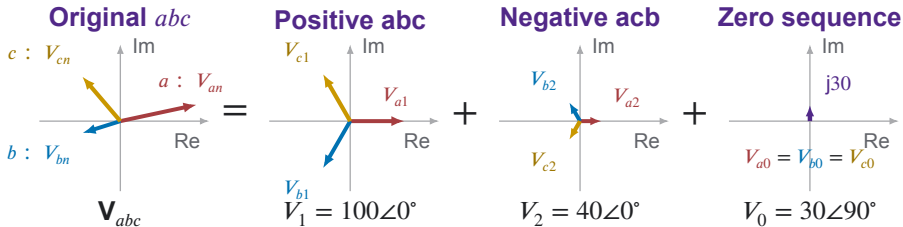
V_0 scales $\mathbf{b}_0$ zero sequence

Reconstruct the three sequence vectors as $V_1 \mathbf{b}_1$, $V_2 \mathbf{b}_2$, and $V_0 \mathbf{b}_0$.

Example: symmetrical components

RMS phasors in volts; all four panels use the same scale:

$$V_{an} = 140 + j30, \quad V_{bn} \approx -70 - j21.962, \quad V_{cn} \approx -70 + j81.962.$$



Verify the reconstruction phase by phase

$$V_{an} = V_{a1} + V_{a2} + V_{a0} = 100 + 40 + j30 = 140 + j30,$$

$$V_{bn} = V_{b1} + V_{b2} + V_{b0} = 100a^2 + 40a + j30 \approx -70 - j21.962,$$

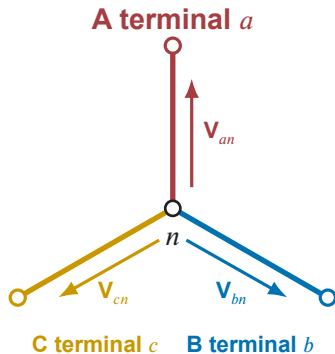
$$V_{cn} = V_{c1} + V_{c2} + V_{c0} = 100a + 40a^2 + j30 \approx -70 + j81.962.$$

PART 3

Wye connection

Line voltage from phase-voltage subtraction

Wye source



Phase voltage

One phase terminal to neutral.

$$V_{an}$$

Line voltage

Potential at terminal a relative to b .

$$V_{ab}$$

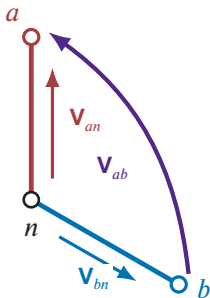
Orientation

The first subscript is the positive reference.

$$V_{xy} = V_x - V_y$$

A line voltage spans two source branches. A phase voltage spans one branch and neutral.

Orientation before arithmetic



Follow the voltage path from b to a through neutral:

$$\mathbf{V}_{ab} = \underbrace{-\mathbf{V}_{bn}}_{b \rightarrow n} + \underbrace{\mathbf{V}_{an}}_{n \rightarrow a}$$

$$\boxed{\mathbf{V}_{ab} = \mathbf{V}_{an} - \mathbf{V}_{bn}}$$

The minus sign comes from reversing the defined direction of $\mathbf{V}_{bn}$.

Wye voltage: complex subtraction

For positive sequence with $V_{an} = V_p \angle 0^\circ$:

Reverse the phase-b voltage

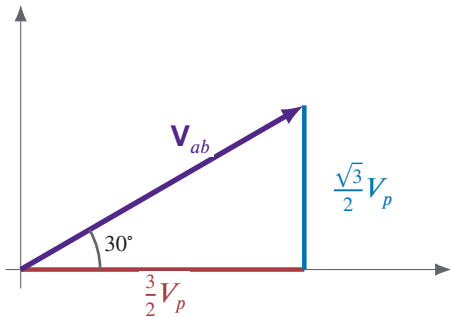
$$V_{bn} = V_p \left(-\frac{1}{2} - j\frac{\sqrt{3}}{2} \right).$$
$$-V_{bn} = V_p \left(\frac{1}{2} + j\frac{\sqrt{3}}{2} \right).$$

Add the phase-a voltage

$$V_{an} = V_p(1 + j0).$$
$$V_{ab} = V_{an} - V_{bn} = V_p \left(\frac{3}{2} + j\frac{\sqrt{3}}{2} \right).$$

Reversing V_{bn} changes both component signs. The resulting real and imaginary parts determine the line voltage.

Component triangle



$$|\mathbf{V}_{ab}| = V_p \sqrt{\frac{9}{4} + \frac{3}{4}} = \sqrt{3} V_p$$

$$\angle \mathbf{V}_{ab} = \arctan\left(\frac{\sqrt{3}/2}{3/2}\right) = 30^\circ.$$

$$\boxed{\mathbf{V}_{ab} = \sqrt{3} V_p \angle 30^\circ}$$

Other two line voltages

A relative to B

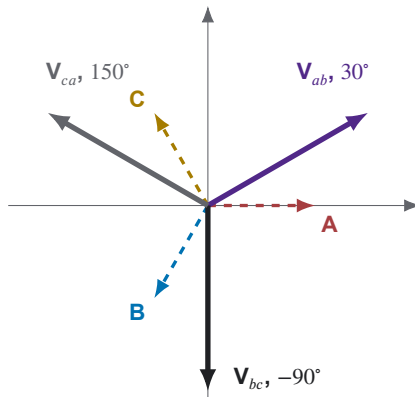
$$\mathbf{V}_{ab} = \mathbf{V}_{an} - \mathbf{V}_{bn} = \sqrt{3}V_p \angle 30^\circ$$

B relative to C

$$\mathbf{V}_{bc} = \mathbf{V}_{bn} - \mathbf{V}_{cn} = \sqrt{3}V_p \angle (-90^\circ)$$

C relative to A

$$\mathbf{V}_{ca} = \mathbf{V}_{cn} - \mathbf{V}_{an} = \sqrt{3}V_p \angle 150^\circ$$



The line voltages remain balanced and keep the same positive sequence.

Source: Overbye, ECEN 460 Lecture 3 (2025), balanced-wye relations. V2 redraw.

Line-to-line rating

Given a positive-sequence line voltage

$$\mathbf{V}_{ab} = V_{LL} \angle \theta_{ab}$$

$$\mathbf{V}_{an} = \frac{V_{LL}}{\sqrt{3}} \angle (\theta_{ab} - 30^\circ)$$

Common reference choice

If $\mathbf{V}_{an} = V_p \angle 0^\circ$, then

$$\mathbf{V}_{ab} = V_{LL} \angle 30^\circ, \quad V_p = \frac{V_{LL}}{\sqrt{3}}.$$

Bus nominal kV is line-to-line. A per-phase circuit uses phase-to-neutral voltage.

Divide the magnitude by $\sqrt{3}$ and subtract 30° from the line-voltage angle.

Example: two voltage reference choices

Given a line phasor

$$\begin{aligned}V_{ab} &= 480\angle 0^\circ \text{ V.} \\V_{an} &= \frac{480}{\sqrt{3}}\angle(0^\circ - 30^\circ) \\&= 277.128\angle(-30^\circ) \text{ V.}\end{aligned}$$

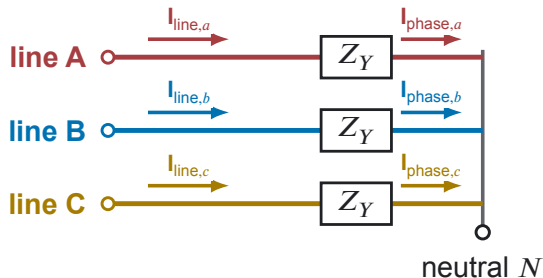
Given only a line-voltage rating

$V_{LL} = 480 \text{ V}$ specifies a magnitude.
Choose $V_{an} = 277.128\angle 0^\circ \text{ V}$.

$$V_{ab} = 480\angle 30^\circ \text{ V.}$$

The two descriptions differ by a common reference rotation. Preserve the given angle whenever the input is a phasor.

Current relation



Each line conductor feeds exactly one load branch before the three paths meet at neutral N .

$$I_{\text{line},a} = I_{\text{phase},a} \equiv I_a,$$

$$I_{\text{line},b} = I_{\text{phase},b} \equiv I_b,$$

$$I_{\text{line},c} = I_{\text{phase},c} \equiv I_c.$$

The continuous current path gives $I_L = I_\phi$ in a wye connection. The $\sqrt{3}$ relation applies to voltage.

Balanced neutral current

Let $a = e^{j2\pi/3}$. For a positive-sequence balanced current set,

$$I_A = I, \quad I_B = a^2 I, \quad I_C = a I,$$

$$I_N = I_A + I_B + I_C = I(1 + a + a^2) = \boxed{0}.$$

Magnitude

equal in all phases

Balance fixes the three current magnitudes.

Angle

successive 120° rotations

The three vectors close into a triangle.

Wye branches share a neutral point. In delta, each branch connects two line terminals.

PART 4

Delta connection

Branch voltage first, then branch and line currents

Delta phase voltage

Each delta branch lies directly between two line terminals. Therefore its phase voltage is a line-to-line voltage:

$$V_{\phi,AB} = V_{AB}, \quad V_{\phi,BC} = V_{BC},$$

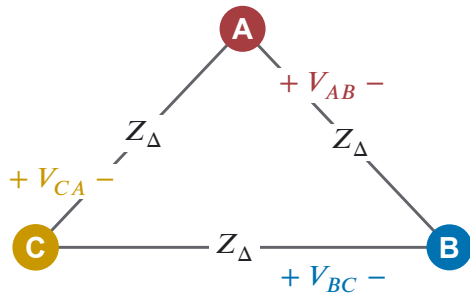
$$V_{\phi,CA} = V_{CA}.$$

For a balanced positive-sequence set, take V_{AB} as the reference:

$$V_{AB} = V_{LL} \angle 0^\circ,$$

$$V_{BC} = V_{LL} \angle (-120^\circ),$$

$$V_{CA} = V_{LL} \angle 120^\circ.$$



In a delta connection, each phase voltage is one of the line-to-line voltages.

Current notation in delta

Single subscript: line current

I_A , I_B , and I_C flow in the external conductors and enter terminals A, B, and C.

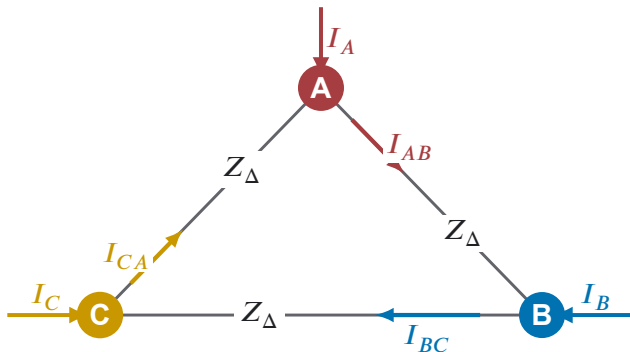
Double subscript: phase current

These currents flow inside the delta branches:

$$I_{AB} : A \rightarrow B,$$

$$I_{BC} : B \rightarrow C,$$

$$I_{CA} : C \rightarrow A.$$



Delta branch currents from voltage

For the main derivation, use three equal resistances $Z_{\Delta} = R_{\Delta}$ and define

$$I_{\phi} = \frac{V_{LL}}{R_{\Delta}}.$$

The branch (phase) currents follow directly from the branch voltages:

$$I_{AB} = \frac{V_{AB}}{R_{\Delta}} = I_{\phi} \angle 0^{\circ},$$

$$I_{BC} = \frac{V_{BC}}{R_{\Delta}} = I_{\phi} \angle (-120^{\circ}),$$

$$I_{CA} = \frac{V_{CA}}{R_{\Delta}} = I_{\phi} \angle 120^{\circ}.$$

With R_{Δ} , each branch current has the same angle as its line-to-line voltage.

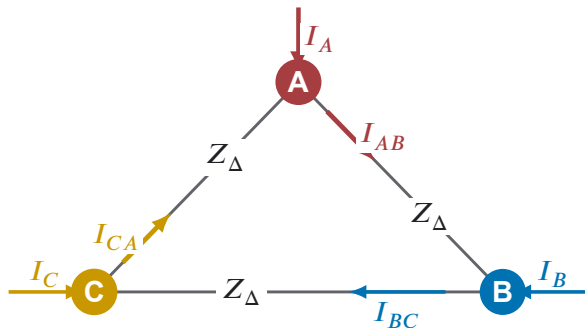
From voltage to line current at terminal A

I_A is the external line current. At terminal A, I_A and I_{CA} enter, while I_{AB} leaves.

$$I_A = I_{AB} - I_{CA}.$$

With $Z_\Delta = R_\Delta$,

$$\begin{aligned} I_A &= \frac{V_{AB}}{R_\Delta} - \frac{V_{CA}}{R_\Delta} \\ &= \boxed{\frac{V_{AB} - V_{CA}}{R_\Delta}}. \end{aligned}$$



The A-line current comes from the difference of the two branch voltages connected to terminal A.

All three line currents from voltage

Apply the same terminal equation at A, B, and C, then replace each branch current by V/R_{Δ} :

$$I_A = I_{AB} - I_{CA} = \frac{V_{AB} - V_{CA}}{R_{\Delta}},$$

$$I_B = I_{BC} - I_{AB} = \frac{V_{BC} - V_{AB}}{R_{\Delta}},$$

$$I_C = I_{CA} - I_{BC} = \frac{V_{CA} - V_{BC}}{R_{\Delta}}.$$

Each external line current is obtained from the two line-to-line voltages that meet at its terminal.

A-line current: algebra from voltage

For positive sequence, $V_{CA} = V_{AB}e^{j2\pi/3}$. Substitute this voltage into the A-terminal result:

$$\begin{aligned} I_A &= \frac{V_{AB} - V_{CA}}{R_\Delta} \\ &= \frac{V_{AB}}{R_\Delta} (1 - e^{j2\pi/3}) \\ &= \frac{V_{AB}}{R_\Delta} \left(\frac{3}{2} - j\frac{\sqrt{3}}{2} \right) \\ &= \boxed{\sqrt{3} \frac{V_{AB}}{R_\Delta} \angle(-30^\circ)} = \boxed{\sqrt{3} I_{AB} \angle(-30^\circ)}. \end{aligned}$$

I_A is $\sqrt{3}$ times the AB branch current and is shifted by -30° .

A-line current: voltage geometry

Geometry of $V_{AB} - V_{CA}$

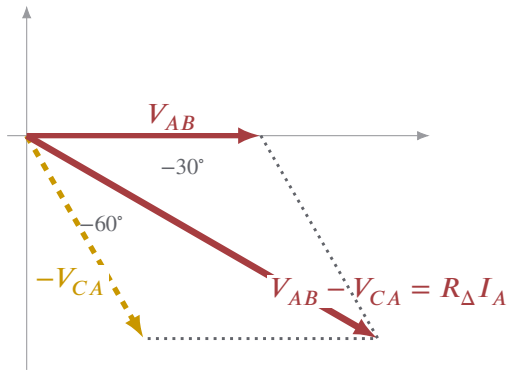
Take V_{AB} as the reference at 0° .

V_{CA} is at 120° , so $-V_{CA}$ is at -60° .

Their sum lies at -30° and has magnitude $\sqrt{3}V_{LL}$.

Dividing by R_Δ changes only the magnitude:

$$I_A = \frac{V_{AB} - V_{CA}}{R_\Delta}.$$



Balanced delta current result

$$I_A = \sqrt{3} I_{AB} \angle -30^\circ$$

The B- and C-terminal voltage equations have the same form, so

$$I_B = \sqrt{3} I_{BC} \angle -30^\circ, \quad I_C = \sqrt{3} I_{CA} \angle -30^\circ.$$

Therefore I_A , I_B , and I_C are also balanced and satisfy

$$I_A + I_B + I_C = 0.$$

Replacing R_Δ by one common complex Z_Δ rotates every branch and line current equally. The $\sqrt{3}$ magnitude ratio and relative -30° shift remain unchanged.

For a balanced delta load, $I_L = \sqrt{3} I_\phi$ and each line current is 30° behind its corresponding branch current.

Positive-sequence delta current phasors

Dashed: branch currents

$$I_{AB} = I_{\phi} \angle 0^{\circ},$$

$$I_{BC} = I_{\phi} \angle (-120^{\circ}),$$

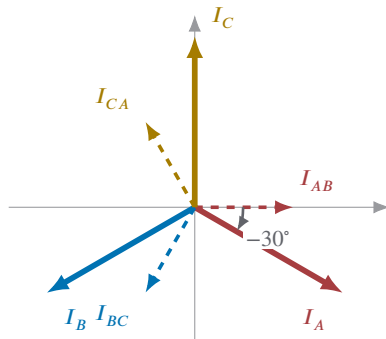
$$I_{CA} = I_{\phi} \angle 120^{\circ}.$$

Solid: line currents

$$I_A = \sqrt{3} I_{\phi} \angle (-30^{\circ}),$$

$$I_B = \sqrt{3} I_{\phi} \angle (-150^{\circ}),$$

$$I_C = \sqrt{3} I_{\phi} \angle 90^{\circ}.$$



Line currents are $\sqrt{3}$ longer and lag the corresponding branch currents by 30° .

Negative-sequence connection shifts

Wye voltage

$V_{bn} = V_{an}e^{j2\pi/3}$ for negative sequence.

$$\begin{aligned} V_{ab} &= V_{an}(1 - e^{j2\pi/3}) \\ &= \boxed{\sqrt{3}V_{an}\angle(-30^\circ)}. \end{aligned}$$

Delta current

$I_{CA} = I_{AB}e^{-j2\pi/3}$ with the same branch arrows.

$$\begin{aligned} I_A &= I_{AB}(1 - e^{-j2\pi/3}) \\ &= \boxed{\sqrt{3}I_{AB}\angle 30^\circ}. \end{aligned}$$

The physical subtraction equations stay the same. Reversing sequence conjugates the connection factor and reverses its 30° angle.

Wye and delta: one reference table

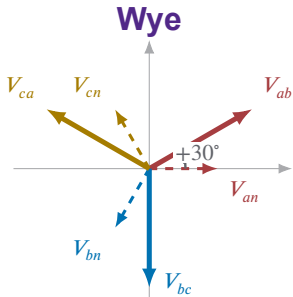
Balanced positive sequence; line currents enter the load:

Relation	Wye	Delta
Voltage magnitude	$V_{LL} = \sqrt{3}V_{\phi}$	$V_{LL} = V_{\phi}$
Current magnitude	$I_L = I_{\phi}$	$I_L = \sqrt{3}I_{\phi}$
Signed phasor relation	$V_{ab} = \sqrt{3}V_{an}\angle 30^{\circ}$	$I_A = \sqrt{3}I_{AB}\angle(-30^{\circ})$
Physical equation	$V_{ab} = V_{an} - V_{bn}$	$I_A = I_{AB} - I_{CA}$

For negative sequence with the same labels and arrows, the 30° signs reverse. Magnitude factors remain the same.

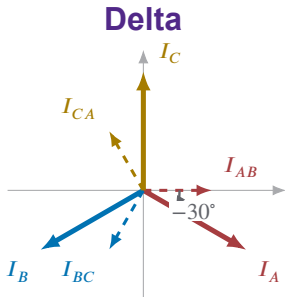
Wye and delta: visual summary

Balanced positive sequence; all vectors within each panel use the same scale.



$$V_{LL} = \sqrt{3}V_{\phi}, \quad +30^\circ$$

$$I_L = I_{\phi}$$



$$V_{LL} = V_{\phi}$$

$$I_L = \sqrt{3}I_{\phi}, \quad -30^\circ$$

Wye creates the $\sqrt{3}$, $+30^\circ$ relation in voltage; delta creates the $\sqrt{3}$, -30° relation in current.

PART 5

Worked example

Complete branch and line currents for a 13.8-kV delta load

Example: balanced 13.8-kV delta

Given and referenced

$$V_{AB} = 13.8\angle 0^\circ \text{ kV}, \quad Z_{\Delta} = 100\angle 20^\circ \Omega.$$

Positive sequence; negligible line impedance.

$$V_{BC} = 13.8\angle(-120^\circ) \text{ kV},$$

$$V_{CA} = 13.8\angle 120^\circ \text{ kV}.$$

Calculate the currents

- Find I_{AB} , I_{BC} , and I_{CA} from V/Z_{Δ} .
- Use terminal KCL to find I_A , I_B , and I_C .
- Verify the line/branch magnitude ratio and angle shift.

$$I_{AB} : A \rightarrow B, \quad I_{BC} : B \rightarrow C,$$

$$I_{CA} : C \rightarrow A.$$

First divide each branch voltage by Z_{Δ} ; then apply the line-current node equations.

Source: Adapted from T. Overbye, ECEN 460, Spring 2025, Lecture 3.

Example: delta branch-current calculation

Each delta branch receives its corresponding line-to-line voltage:

$$I_{AB} = \frac{13,800\angle 0^\circ}{100\angle 20^\circ} = 138\angle(-20^\circ) \text{ A},$$

$$I_{BC} = \frac{13,800\angle(-120^\circ)}{100\angle 20^\circ} = 138\angle(-140^\circ) \text{ A},$$

$$I_{CA} = \frac{13,800\angle 120^\circ}{100\angle 20^\circ} = 138\angle 100^\circ \text{ A}.$$

Divide magnitudes and subtract angles. All branch currents lag their own branch voltages by the impedance angle, 20° .

Example: line current from rectangular KCL

At terminal A, $I_A = I_{AB} - I_{CA}$.

$$I_{AB} = 129.678 - j47.199 \text{ A}, \quad I_{CA} = -23.963 + j135.903 \text{ A}.$$

$$\begin{aligned} I_A &= (129.678 - j47.199) - (-23.963 + j135.903) \\ &= 153.641 - j183.102 \text{ A} \quad (\text{using unrounded inputs}) \\ &= \boxed{239.023 \angle (-50^\circ) \text{ A}}. \end{aligned}$$

$$\sqrt{3}I_{AB} \angle (-30^\circ) = \sqrt{3}(138) \angle (-20^\circ - 30^\circ).$$

Direct KCL and the balanced connection factor produce the same line current.

Example: branch and line current sets

$$I_{AB} = \frac{13,800 \angle 0^\circ}{100 \angle 20^\circ} = 138 \angle (-20^\circ) \text{ A.}$$

Terminal	Branch current (A)	Line current (A)
A	$I_{AB} = 138 \angle (-20^\circ)$	$I_A = 239.023 \angle (-50^\circ)$
B	$I_{BC} = 138 \angle (-140^\circ)$	$I_B = 239.023 \angle (-170^\circ)$
C	$I_{CA} = 138 \angle 100^\circ$	$I_C = 239.023 \angle 70^\circ$

$$I_A = I_{AB} - I_{CA} = \sqrt{3} I_{AB} \angle (-30^\circ), \quad I_A + I_B + I_C = 0.$$

Each branch current lags its own voltage by 20° . The line current adds a separate -30° connection shift.